

Hormones

CHEMICAL MESSAGES CALLED HORMONES CONTROL DAY-TO-DAY BODY PROCESSES.

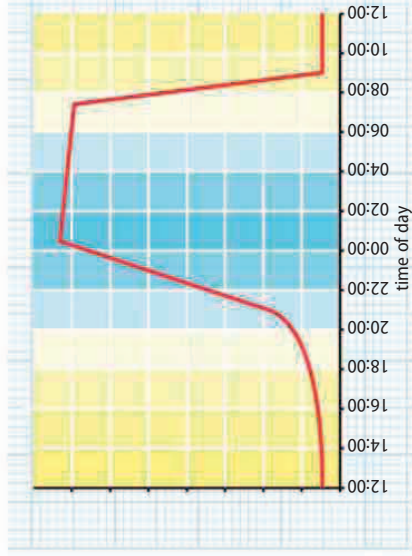
Complex life forms use hormones to control growth, metabolic rate, and to prepare the body for activity or sleep. Hormones are produced in special organs called glands throughout the body.

Glands

Any body part that secretes a substance is called a gland. Exocrine glands send chemicals out of the body. They include sweat, salivary, and the seminal gland, which releases semen. Hormones are produced by endocrine glands, which release substances into the blood and internal body fluids. From there, hormones are carried to the parts of the body that they influence.

▷ Melatonin

This hormone is released by the pineal gland underneath the brain. Its production is linked to the time of the day. In humans, it is released in the evening to prepare the body for night time, making us sleepy. In nocturnal animals, it is activated to wake them up.



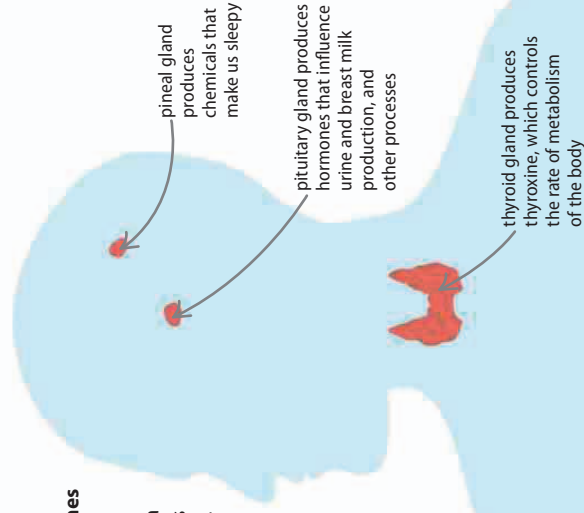
▷ Epinephrine

This powerful hormone is released by the adrenal glands. It triggers the body's response to stress (the "fight-or-flight" response). When released, epinephrine (also known as adrenaline) gives an immediate energy boost to prepare the body to act. Some of the common signs of this are listed here.

Effect	Explanation	Purpose
skin goes pale	blood vessels in the skin contract	blood directed to muscles for movement
heart rate goes up	heart pumps more blood	oxygen reaches muscles faster
heavy breathing	lungs take bigger breaths	boost to oxygen supply

▷ Human hormones

The glands shown here secrete hormones used in a variety of processes in the human body.



adrenal glands produce epinephrine to combat stress

pancreas produces insulin to process sugar in food

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- 《 36-37 Transport systems
- 《 38-39 Movement

Radiating heat